NOVEMBER 2011 EXAMINATION

Nelson Mandela Metropolitan University

for tomorrow

MODULE DESCRIPTION	:	**Advanced Programming 3.2**
MODULE CODE	:	**WRAP302**
FACULTY	:	**Science**
QUALIFICATION	:	**BSc, BCom**
EXAMINATION DATE	:	**21 November 2011**
SESSION	:	**14:00**
DURATION (IN MINUTES)	:	**180**
TOTAL MARKS	:	**100**
PAGES	:	**4**
ADDENDUMS	:	**None**
EXAMINER	:	**Dr D. Vogts**
MODERATOR	:	**Prof G.C. Wells (Rhodes)**

INSTRUCTIONS :

1. **Answer ALL questions.**
2. **No permitted literature for Section A (written).**
3. **Submit Section A to receive Section B (practical).**
4. **Java textbook and/or printed lecture notes allowed for Section B.**

DO NOT TURN THE PAGE BEFORE TOLD TO DO SO

Section A: Written [25]

Note: This section is closed book. You may not make use of any text books or lecture notes.

1) The custom implementation class `StatisticKeyMap` implements the `SortedMap` interface.

1.1) What kind of functionality would you expect the `StatisticKeyMap` class to implement simply by knowing that it implements the `SortedMap` interface? (4)

1.2) In order for the `StatisticKeyMap` to be sorted, the class makes use of a `Comparator`. What is the purpose of the comparator? (2)

2.1) A thread may be in a number of different states during its lifetime. Draw a diagram that shows the states, as well as what causes the thread to changes states. Clearly label all states and transitions. (5)

2.2) Why is it sometimes (but not always) necessary to use the `synchronized` keyword when writing code that you want to be thread safe? (2)

2.3) Synchronizing methods can also lead to problems of their own, such as *dead lock*. Explain what dead lock is, using an example to illustrate your explanation. (3)

3.1) You have probably noticed that sometimes when loading images, it takes a while before they can be drawn correctly onscreen. Explain briefly how you can ensure that all images have been loaded before you start using them. (2)

3.2) Briefly discuss the process required when wanting to use the Media Player to load a video and play it onscreen. (7)

Hand in your answer book to obtain the next section.

Section B: Practical [75]

Note: this section is open book. You may use a Java text book and/or printed lecture notes.

Use the username and password provided to log on.

Username: _____ Password: _____

1) During the first few weeks at University, those students that are enrolled for the computer literacy modules (WRFC101, WRFC141, etc.) need to go through a scheduling procedure in order to be assigned into specific slots for the lectures, lessons and applications presented in the module. For each session type, there are a number of duplicates and the student needs to pick from these such that it does not clash with any of their other modules. (75)

A session has a number of pieces of information that describe it. These include:

- the module code, e.g. "WRFC101";
- the type of session, e.g. "lecture", "lesson" or "application";
- the time slot it occupies on the lecture time table, e.g. "Tu1", "We3+We4", etc.;
- the maximum number of students who can attend that session; as well as
- a list of student numbers of those students who are scheduled for that particular session.

You need to write a set of console-based client and server applications that allow students to book their sessions. The server must function in the following manner:

- on start up, the server must start listening to port 5050;
- the server can handle a number of simultaneous client connections;
- the server initialises the sessions shown below;

Module Code	Type	Timetable Slots	Capacity
WRFC101	lecture	Tu1+Tu2	5
WRFC101	lecture	Tu3+Tu4	5
WRFC101	lesson	We3+We4	3
WRFC101	lesson	Th5+Th6	4
WRFC101	application	Th1+Th2	10
WRFC101	application	Th7+Th8	15
WRFC141	lecture	Mo1+Mo2	5
WRFC141	lecture	Th3+Th4	10
WRFC141	lesson	Tu4+Tu5	2
WRFC141	lesson	Th5+Th6	3
WRFC141	application	We1+We2	5
WRFC141	application	Th6+Th7	10

- the server responds to commands from a client;
- standard DataInputStreams and DataOutputStreams are used for communication to

allow the server to be more flexible.

The client must function in the following manner:

- when the client starts, it connects to the server;
- the user is asked for their student number;
- the client asks the user for commands continuously and processes them until the user enters the command “DISCONNECT” which disconnects the client from the server;
- the client can issue the following commands:
    - “QUERY SESSIONS”
        - the command has parameters for the module code and the type of session.
        - the command and parameters are sent to the server, which compiles a list of all the available timetable slots for that particular module and type of session. Only timetable slots that are not filled to capacity are returned.
        - the list is returned by the server and displayed on the screen.
        - for example, if the server was sent “QUERY SESSIONS”, “WRFC101” and “lecture”, it would return a list containing “Tu1+Tu2” and “Tu3+Tu4” if both slots were not filled to capacity.
    - “BOOK”
        - the command has parameters for the module code, the type of session, timetable slot and the student number of the user.
        - the command and parameters are sent to the server, which attempts to place the student into the list of students allocated to the session. The student will not be added if he/she has already been allocated to the session or the session is full.
        - the server returns a true/false value indicating whether the student could be added or not.
    - “UNBOOK”
        - the command has parameters for the module code, the type of session, timetable slot and the student number of the user.
        - the command and parameters are sent to the server, which attempts to remove the student from the list of students allocated to the session. The student will only be removed from the list if they were allocated to it.
- appropriate messages must be displayed by the client to indicate to the user what is happening.

Important: you must make use of appropriate Java Collection Framework classes when implementing your solution!

Important: you must ensure that the data cannot be corrupted due to race-conditions!

To get a feeling of how the application should function, you may run the runServer.bat and runClient.bat files found on the examination folder.

End of Exam Paper